

L7 – Trig Identities

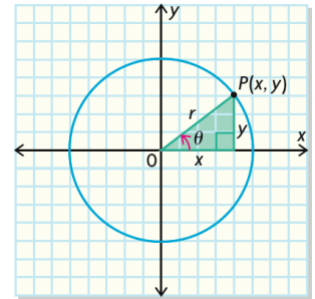
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_____ : A mathematical equation that is true for ALL values of the given variables.

Part 1: Proving the Pythagorean and Quotient Identities

For this part you will need to remember that trig ratios can be written in terms of x and y

Example 1: Prove the quotient identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$



Example 2: Prove the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$

Fundamental Trigonometric Identities		
Reciprocal Identities	Quotient Identities	Pythagorean Identities
$\csc \theta = \frac{1}{\sin \theta}$ $\sec \theta = \frac{1}{\cos \theta}$ $\cot \theta = \frac{1}{\tan \theta}$	$\frac{\sin \theta}{\cos \theta} = \tan \theta$ $\frac{\cos \theta}{\sin \theta} = \cot \theta$	$\sin^2 \theta + \cos^2 \theta = 1$

Tips and Tricks		
Reciprocal Identities	Quotient Identities	Pythagorean Identities
Square both sides $\csc^2 \theta = \frac{1}{\sin^2 \theta}$ $\sec^2 \theta = \frac{1}{\cos^2 \theta}$ $\cot^2 \theta = \frac{1}{\tan^2 \theta}$	Square both sides $\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$ $\frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta$	Rearrange the identity $\sin^2 \theta = 1 - \cos^2 \theta$ $\cos^2 \theta = 1 - \sin^2 \theta$
General tips for proving identities: i) Try to change everything to $\sin \theta$ or $\cos \theta$ ii) If you have to fractions being added or subtracted, find a common denominator and combine the fractions iii) Use difference of squares $\rightarrow 1 - \sin^2 \theta = (1 - \sin \theta)(1 + \sin \theta)$ iv) Use the power rule $\rightarrow \sin^6 \theta = (\sin^2 \theta)^3$		

We will use the preceding identities to help us prove more complex identities in the following examples.

Example 3: Prove each of the following identities

a) $\frac{\cos \theta \tan \theta}{\sin \theta} = 1$

b) $\tan^2 \theta + 1 = \sec^2 \theta$

c) $\cos^2 x = (1 - \sin x)(1 + \sin x)$

$$\mathbf{d)} \frac{\sin^2 x}{1 - \cos x} = 1 + \cos x$$

$$\mathbf{e)} \sin \theta \sec \theta \cot \theta = 1$$

$$\mathbf{f)} \frac{1}{1 - \sin x} - \frac{1}{1 + \sin x} = \frac{2 \tan x}{\cos x}$$

g) $(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = 2$

h) $\tan x + \frac{\cos x}{1 + \sin x} = \sec x$